



Society of Petroleum Engineers

SPE-230176-MS

Clamp-On Real-Time Wellhead Monitoring for Flow Behavior Characterization

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

This paper focuses on the use of real-time, clamp-on Electromagnetic Acoustic Transducer (EMAT) measurements at the wellhead to directly monitor real-time, actual well behavior. By capturing high-resolution flow data upstream of separation, we can observe how the well responds dynamically to different operational scenarios, particularly when routed to test separators versus flowing into the bulk line. These real-time measurements reveal systematic changes in flow regime, production rate, and transient behavior depending on downstream backpressure and flow path configuration. Understanding these variations allows for a more accurate assessment of actual well performance, independent of separator lag or test facility constraints, and supports improved production optimization in unconventional assets.

The EMAT clamp-on sensor was deployed at the wellhead to capture real-time multiphase flow data across various operational states, with a focus on transitions between test separator routing and flow into the bulk line. Sensor configuration accounted for pipe geometry, fluid properties (PVT), and historical production data to ensure accurate signal interpretation. Measurements were conducted across multiple wells and compared against downstream test data. Particular attention was given to identifying systematic changes in flow dynamics such as shifts in phase behavior, transient responses, and production rates associated with varying the test configuration of the well. These observations were analyzed using multiphase flow models (e.g., Taitel-Dukler, TUFFP) to contextualize real-time signals within expected flow regimes and diagnose operational influences on observed well performance.

Real-time wellhead data captured by the EMAT system revealed consistent and measurable changes in well behavior depending on whether the well flow was directed to a test separator or the bulk line. Transient analysis showed distinct shifts in flow regime during routing transitions, highlighting the impact of downstream backpressure and infrastructure configuration on apparent well performance. These differences were not immediately evident in downstream measurements due to system lag, averaging effects and limitations of conventional separator measurements. This capability supports improved lift performance assessment, better production allocation, and more responsive optimization strategies in dynamic field environments.

This work demonstrates the unique value of real-time, non-intrusive wellhead measurements in uncovering the true behavior of unconventional wells under varying operational conditions. The system

can be also deployed on conventional wells. By capturing transient flow dynamics independent of downstream infrastructure effects, the EMAT system provides a more accurate, high-resolution view of well performance. This capability enables operators to detect and interpret systematic changes when transitioning between test and bulk flow, allowing for more precise diagnostics, improved production allocation, and faster response to evolving well conditions. The approach offers a practical and scalable method to enhance surveillance and optimization strategies across diverse field settings.

Introduction

This paper presents field-based measurements acquired on a gas lift well in the Bakken formation using a novel clamp-on Multiphase Flow Sensor. Building on previous deployments of the EMAT-based clamp-on flow sensor (Black et al. 2023a, 2023b, 2024a, 2024b), the study focuses on the application of real-time flow regime identification and classification to optimize gas lift performance. The multiphase flow sensing technology enables continuous, non-intrusive monitoring of multiphase flow, offering a scalable, low-cost alternative to traditional well testing methods. By integrating real-time flow regime data with field operations, this approach supports more dynamic and informed gas lift optimization, enhancing both production efficiency and reservoir insight.

Traditionally in Bakken gas lift operations, periodic well testing is typically performed using shared separator infrastructure, which captures production data from a well for only brief intervals. These short-duration tests may fail to identify critical instabilities or flow regime transitions that can persist over longer timescales. Especially in gas lift wells, where transient multiphase flow behavior can vary significantly over hours or days, reliance on intermittent data can lead to suboptimal injection strategies, deferred production, and misleading conclusions about well performance.

Clamp-on multiphase flow sensing systems overcome these limitations by providing continuous, high-frequency measurements of oil, gas, and water flow rates, along with dynamic insights into the prevailing flow regime (e.g., stratified, slug, or annular). This allows operators to detect shifts in flow behavior associated with gas lift inefficiency, unstable lift conditions, or liquid loading. In this study, flow regime classification was used as a diagnostic tool to guide lift gas adjustment and identify opportunities for intervention without interrupting production or requiring test separator access.

Deploying these compact, non-invasive meters enables real-time data acquisition across multiple wells simultaneously, reducing both capital and operational expenditure. Furthermore, by streaming data to centralized cloud-based platforms, field-wide flow behavior can be monitored holistically, revealing spatial patterns and reservoir-scale insights that support coordinated lift optimization across the asset. This cooperative measurement approach offers a marked improvement over isolated, test-based workflows and aligns with broader goals of digital oilfield transformation.

Fig. 1 illustrates how real-time flow regime data across multiple Bakken gas lift wells, where a clamp-on system is installed at every wellhead, can feed directly into operational intelligence systems, enabling active surveillance and automated lift control. Rather than waiting for periodic test results, operators gain immediate visibility into well health and lift performance, supporting faster and more precise adjustments to maximize drawdown, reduce liquid holdup, and improve overall reservoir recovery.

By integrating continuous flow regime monitoring into gas lift operations, operators can identify and correct unstable flow patterns like slugging or churn flows that waste energy and lower lift efficiency. Previously, detecting rapid pressure fluctuations or gas bypassing in gas-lift wells was challenging using only separator measurements, due to their limited temporal resolution and inability to capture transient flow behaviors. By adjusting how much gas is injected, operators can improve the balance with production, helping lift more liquid with less gas. This real-time control minimizes inefficiencies and boosts production while cutting OPEX.

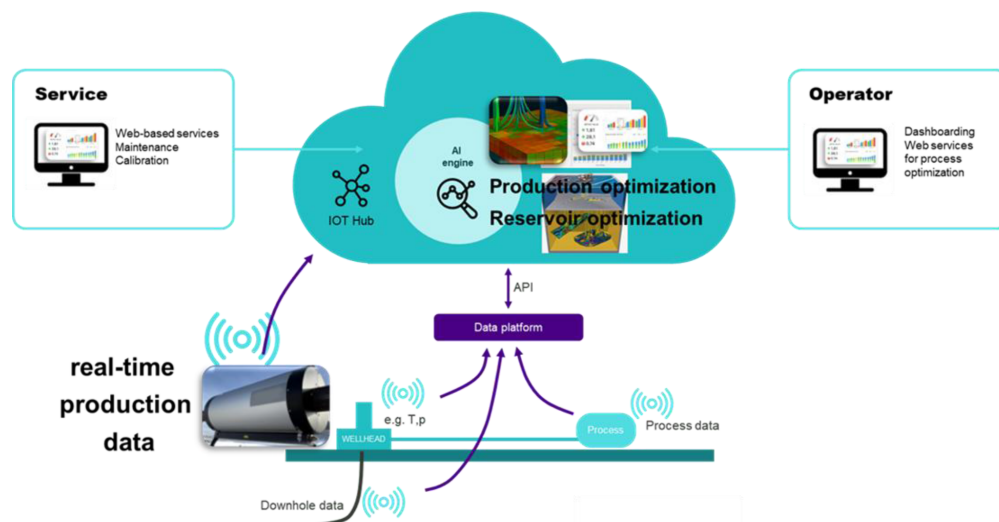


Figure 1—How Continuous Real-Time Multiphase Data Influences Upstream Optimization – the installation of systems at multiple wellheads enables real-time continuous data for every well.

Statement of Theory, Definitions and Measurement Approach

Well Simulation

In the results presented in this paper, an unconventional gas lift well was measured. The well configuration for this is shown schematically in Fig 2(a) and the location of the meter within the surface manifold infrastructure is shown in Fig 2(b). At the well pad where the well is located, well testing is performed through a bulk and test configuration: several wells are shared between a test separator so that well testing takes place in turn between each of the wells. This means the separator only monitors the wells for a small percentage of the time. The clamp-on meter was deployed at the wellhead and therefore the produced flow patterns were monitored continuously, including most of the time when the well was not being measured by the test separator.

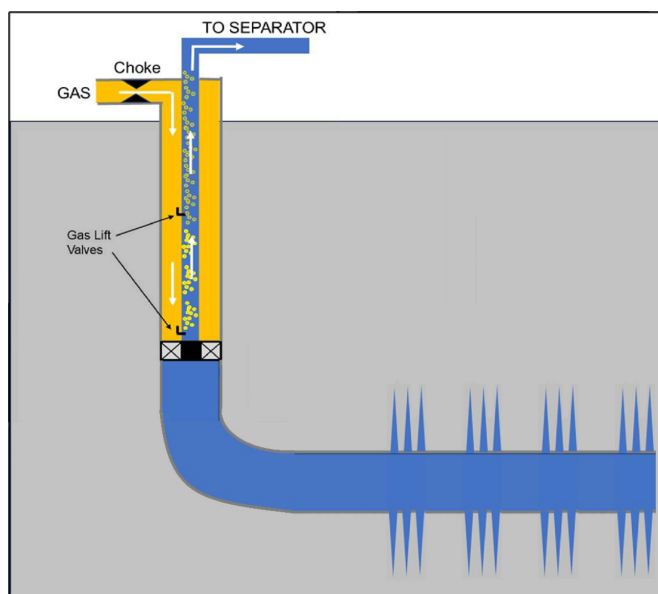


Figure 2a—Schematic diagram showing a horizontal gas lift well. For the present study, we installed our sensor directly onto the production line coming out of the wellhead (see Fig. 4)

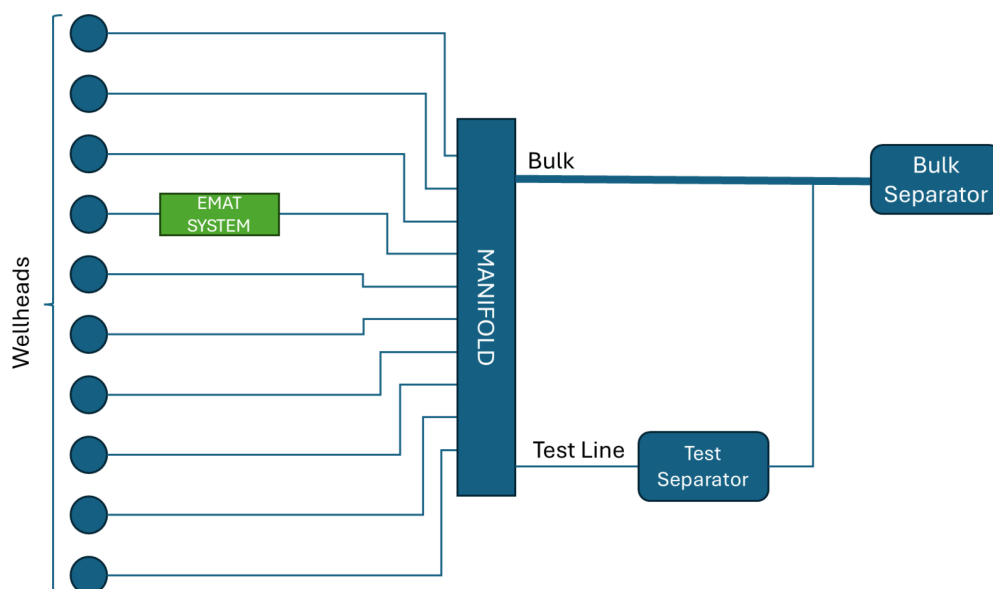


Figure 2b—Schematic diagram showing position of EMAT flow sensor close to the wellhead of one of the gas lift wells on the pad. By default, the production flows into the bulk separator, but when the well is under test, the production from the well is diverted through the test separator.

Prior to performing a measurement, it was first necessary to perform a simulation of the expected flow conditions encountered within the pipe at the point of measurement. To do this, it is necessary to obtain production data so that the expected flow regime and in-pipe flow conditions can be simulated (Taitel and Dukler 1976; Al Safran and Brill 2017). Production data obtained through test separator measurements for the gas lift well are shown in Table 1.

Table 1—Production data for gas lift well. Production rates are derived from publicly available data for this well for the month of April 2025. Line pressure and temperature are the median pressure and temperature in the same period, determined from operator-provided data.

Type of Well	Gas Lift
Pipe Diameter/Schedule	3 in. / Schedule 80
Oil Gravity	39 °API
Brine Salinity	200,000 ppm
Oil Rate	179 B/D
Water Rate	201 B/D
Gas Rate	722 Mscf/D
Line Pressure	121 psi
Line Temperature	108°F

The production data from Table 1 can be used with the Taitel and Dukler model to predict the expected flow regimes for the point of measurement. The results from this are shown in Fig. 3, and the detailed approach to how the calculation is performed has been outlined previously by the authors (Black et al. 2022).

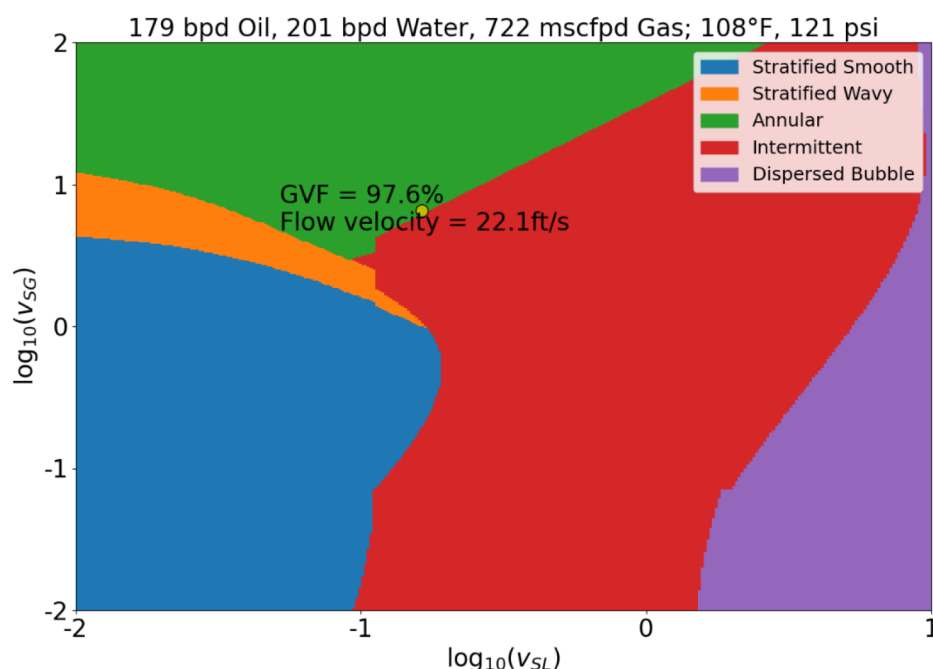


Figure 3—Expected flow regime for the gas lift well, showing it falls within the transition region between intermittent (slugging) and annular flow, based on input conditions.

The simulation plot in Fig. 3 shows the type of flow likely to occur inside the pipe. However, because the input values are averaged over a relatively long timeframe, the plot should be considered only as a rough indicator. In reality, the flow regime in a horizontal well is rarely stable and can switch between annular, intermittent, and stratified smooth or wavy flows as gas and liquid velocities fluctuate. When gas velocity increases, the flow may shift toward fully developed annular flow; when it decreases, intermittent flow can dominate. These transitions happen rapidly and can significantly affect pressure drop, liquid holdup, and artificial lift performance, highlighting the importance of in-situ flow monitoring.

Measurement with Clamp-on Multiphase Flow Sensor

Measurements at the gas lift well were performed using a clamp-on multiphase flowmeter that uses the principle of Electromagnetic Acoustic Transducers (EMAT) to measure multiphase flow. A typical field set up is shown in Fig. 4. The EMAT sensor assembly is seen clamping around a 3-inch Schedule 80 pipe with cable bundles connecting to the processing unit on the ground. A full description of the details of the measurements is outlined previously by the authors (Black et al. 2022). Measurements were performed as shown in Fig. 1, and data was transmitted from the well site using a satellite internet connection.

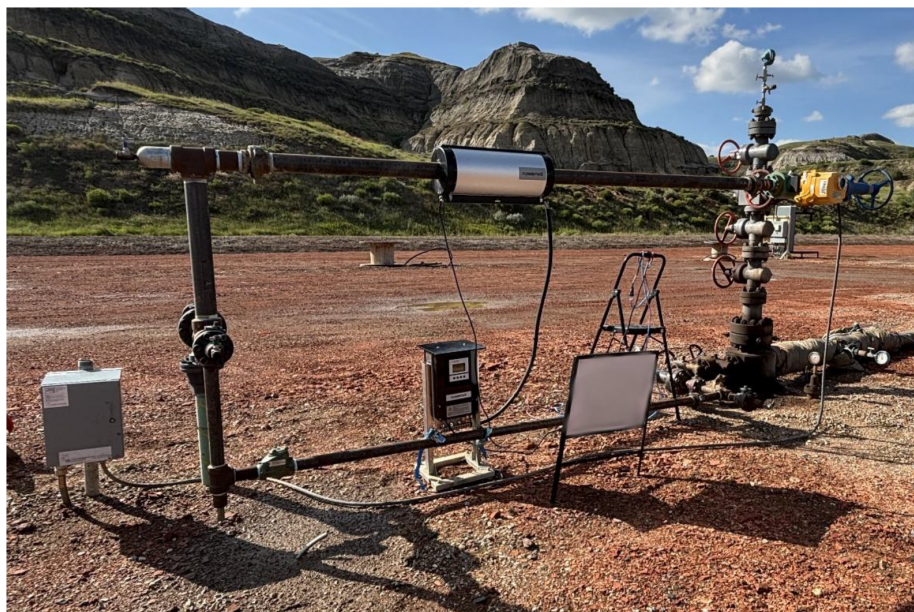


Figure 4—Installation of our flow measurement system in the field.

Results

Liquid and Gas Flows

In this section, we present the measured liquid and gas flow rates acquired using the clamp-on EMAT multiphase flow sensor installed on the gas lift well. These results provide key insight into the well's production performance and serve as the basis for understanding the relationship between flow rates and changing flow regimes. The sensor does have the capability to measure water cut, but for flow regime identification the most relevant parameters are liquid flow and gas flow (Black et al. 2022 – 2024).

Fig. 5 shows a three-month time series of instantaneous liquid and gas flow rates measured using the clamp-on EMAT multiphase flow sensor. Liquid flow rates generally range between 200 and 400 B/D, with occasional spikes up to 600 B/D, while gas rates range from 1,000 to 1,400 Mscf/D. The measured liquid rates align well with the operator's reported figures. However, the measured gas rates are significantly higher because our wellhead-mounted sensor also captures the injected lift gas returning to the surface.

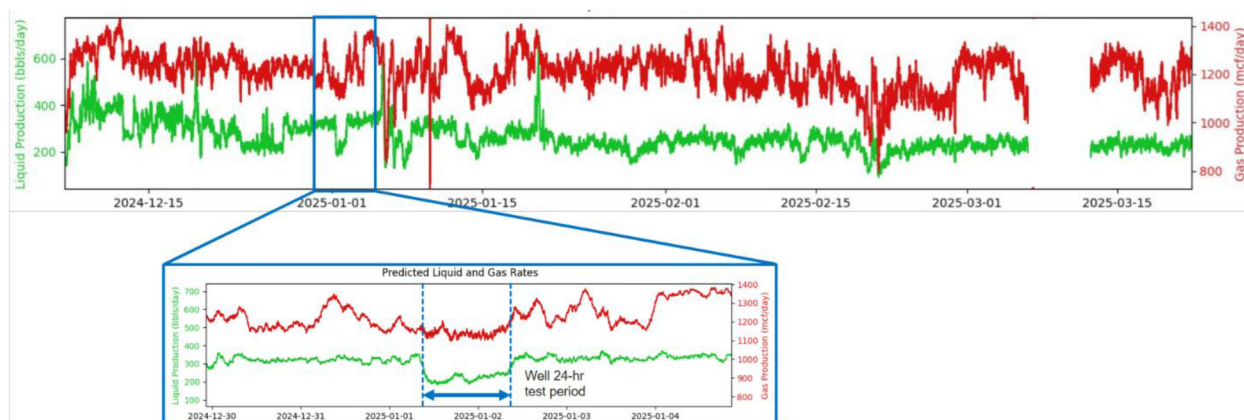


Figure 5—Measured liquid (green) and gas (red) flow rates from the gas lift well using the clamp-on EMAT multiphase flow sensor. The zoomed-in inset shows a week of measurement, including a 24-hour period when the well was routed to the test separator (blue double-ended arrow) instead of the bulk separator. We assume that this change in routing caused a different backpressure on the well, resulting in a noticeably lower total liquid rate compared to the periods before and after.

The inset shows a one-week production period, highlighting a 24-hour interval when the well was routed to the test separator instead of the bulk separator. During this time, the gas flow rate stabilizes at the lower end of its range and the liquid rate drops noticeably. This behavior is consistent with an increase in backpressure caused by the test separator operating at a higher pressure than the bulk system, which slightly chokes the well and reduces production. Cumulative daily rates from the separator when the well was under test were in agreement with the meter values in-line with accuracy levels required for well testing. However, care should be taken to interpret this, because of the systematic influence the separator appears to show.

This highlights a key limitation of traditional well tests: they offer only brief snapshots under conditions that may not match normal operating pressures, in this case underestimating true well production. By contrast, continuous real-time flow measurements reveal transient flow dynamics and provide a more accurate view of well performance, enabling more informed production decisions.

Finally, while these measurements offer valuable insights into overall production, fully optimizing gas lift wells requires understanding the underlying multiphase flow patterns—such as stratified, slug, or annular flow. These patterns directly impact lift efficiency and can inform adjustments to gas injection strategies. The next section explores these flow regimes in detail and examines their effect on well performance.

Flow Patterns

Understanding the underlying flow patterns is key to interpreting the flow data and explaining how the well's production behavior evolves over time. Fig. 6 illustrates how the clamp-on sensor system is positioned around the pipe to detect flow signals for flow pattern analysis.

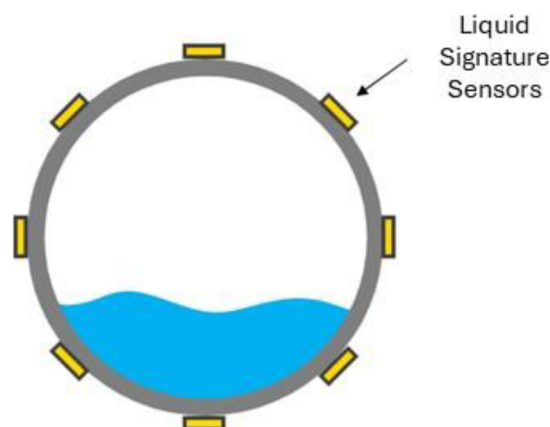


Figure 6—Schematic showing how the clamp-on sensor system is positioned around the pipe to detect transient flow signals for flow pattern analysis where liquid is contacting at different positions around the circumference.

In the context of flow characterization, we distinguish between flow regime and flow pattern. The flow regime describes the instantaneous behavior of the liquid and gas phases, while the flow pattern refers to the cyclic behavior of the wellbore flow, which can alternate between multiple regimes. The sensor system detects transient liquid and gas signatures at different circumferential positions. When sampled at sufficiently high frequency, these measurements capture detailed transient behaviors that reveal how flow regimes develop and change over time.

Fig. 7 shows a classification of the different flow regimes based on the data from the sensors configured as described in Fig. 6. The flow regime analysis demonstrates that the well flow behavior is far more complex than what can be inferred from flow rates alone. As shown in Fig. 7, we typically observe more than ten distinct flow regime transitions within just one hour. These rapid changes highlight dynamic conditions—such as intermittent slugging, stratified/dispersed wavy and annular flow—that are not evident from average flow values, but can have a significant impact on production performance and artificial lift operation.

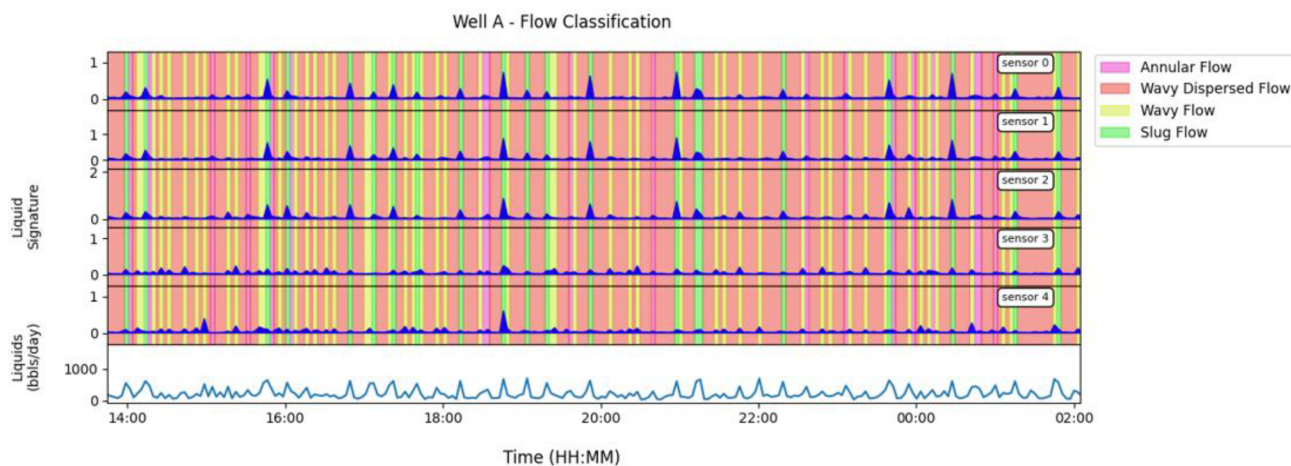


Figure 7—Liquid signatures from five sensors positioned around one-half circumference from the top (Sensor 0) to the bottom (Sensor 4) over a 12-hour period. The other sensors (Sensors 1-3) are positioned at intermediate locations with Sensor 2 positioned at the side of the pipe. Analysis of the flow regime shows that the well cycles between different flow regimes, predominantly slug flow, wavy flow, and gas flow. The bottom panel shows the liquid flow rate corresponding to the liquid signatures above.

Fig. 8 shows high-resolution data of four different flow regimes that we were able to observe. Each sub-figure is divided into two parts, with the top part showing the liquid loading (i.e. liquid holdup) in the pipe, and the bottom part showing how much liquid is touching the inside of the pipe at the position of the sensors placed around the pipe. Liquid hold-up can be calculated by combining the individual sensor behaviors at different positions around the circumference of the pipe and converting the radial liquid coverage into a cross-sectional area which represents the liquid hold-up.

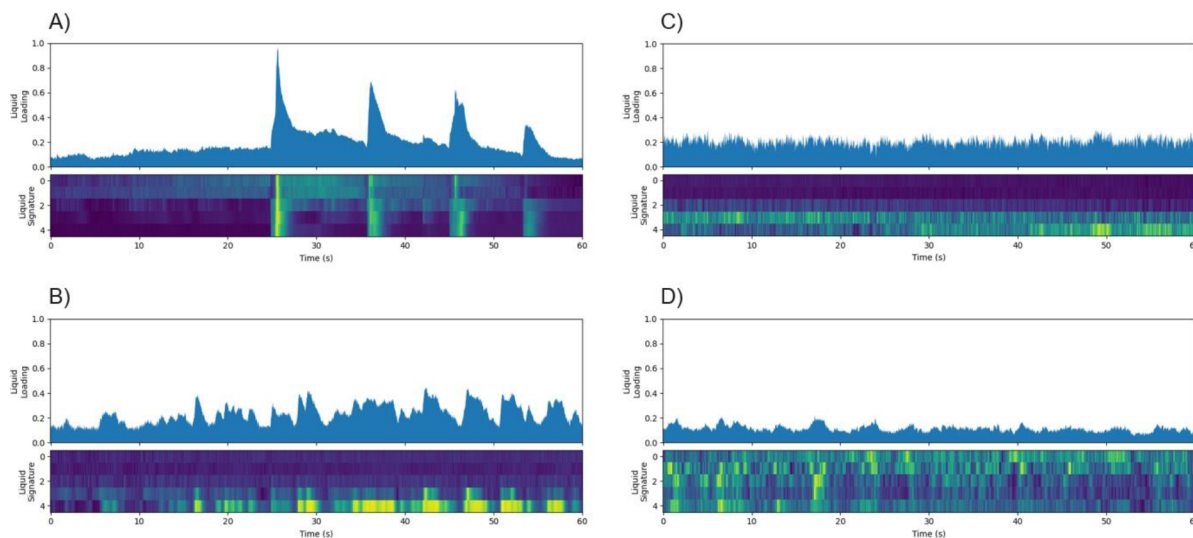


Figure 8—High-resolution examples of slug flow (A), stratified wavy flow (B), wavy dispersed flow (C) and annular flow (D). In slug flow, brief liquid signatures are detected at the top sensor. In wavy flows, liquid signatures are seen on the bottom sensors. In annular flow, chaotic liquid signatures are seen on all sensors.

Fig. 8 A shows relatively low liquid holdup throughout, apart from four large spikes in the liquid holdup observed at 25s, 36s, 46s and 54s. This is characteristic of intermittent flow. In Fig. 8B, we see most of the liquid in the bottom portion of the pipe, with marked oscillations starting around 15s into the measurement. The fact that we see barely any liquid in the top part of the pipe characterizes this type of flow as stratified wavy flow. In Fig. 8C, the liquid signatures are mostly limited to the bottom as well, but there are no significant large waves of liquid. This type of behavior is wavy dispersed flow, where most of the liquid is

coating the bottom of the pipe, and some of the liquid is entrained in the gas in the top of the pipe. If the gas rate increases further, the flow regime transitions to annular flow, which is shown in Fig. 8D, where there are relatively weak liquid signatures at all positions around the circumference of the pipe.

The observed flow regimes have direct implications for gas lift efficiency. For example, the intermittent slugs seen in Fig. 8A indicate unstable liquid transport, which can lead to inconsistent lift performance and increased backpressure, reducing lift efficiency. The stratified wavy flow in Fig. 8B shows that liquid is not fully mobilized, with much of it settling at the bottom and requiring additional gas energy to maintain flow continuity. In Fig. 8C, the wavy dispersed flow shows a mix of liquid coating the bottom and some liquid droplets entrained in the gas core. This partial entrainment is generally more stable than fully stratified flow but still less stable than fully developed annular flow. Finally, Fig. 8D illustrates annular flow, where liquid forms a continuous film around the pipe wall and is consistently carried by the high-velocity gas core. Annular flow represents a more stable condition that minimizes liquid holdup and promotes steady transport. However, achieving and maintaining this regime typically requires higher gas injection rates to keep the liquid film mobilized, which increases compression or fuel costs. Therefore, the operational goal is not to maximize gas injection at all times but to balance injection rates to suppress severe slugging and stratification while avoiding unnecessary lift gas consumption.

Real-time identification of these regimes can support this by enabling dynamic adjustments to gas injection rates. By integrating flow regime detection into a closed-loop control system, operators can potentially optimize lift performance in real time – ensuring uniform liquid removal, reducing pressure surges, and improving the overall efficiency and stability of the artificial lift system.

Conclusion

This work demonstrates that our EMAT-based system can measure liquid and gas flow rates in gas lift wells with good agreement to reference separator data. Beyond verifying rates, the system's capability to visualize flow patterns offers unique insight into how multiphase flow behaves at the wellhead which can give a more direct understanding of downhole conditions: information that is difficult to obtain with conventional separator measurement. This flow pattern visualization is a relatively unique feature of our sensor technology and represents a significant advantage for understanding and optimizing artificial lift performance. To build on these results, further experiments should quantify how specific flow patterns influence lift efficiency, providing actionable insights that help operators adjust gas lift operation to maximize production and optimize lift performance.

Acknowledgements

The authors would like to thank the operator for supporting Rosenxt's efforts and providing access to the Williston Basin wells investigated in this paper.

The authors also acknowledge the work of the members of the Rosenxt Upstream Optimization Team for developing the technology discussed in this paper.

Abbreviations

CAPEX	Capital Expenditure
DUT	Device Under Test
EMAT	Electromagnetic Acoustic Transducer
GVF	Gas Volume Fraction
MPFM	Multiphase Flowmeter
OPEX	Operating Expenditure
RPM	Revolutions Per Minute
TUFFP	Tulsa University Fluid Flow Projects

WLR Water/Liquid Ratio

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